MIL-C-36625
7 January 1970

MILITARY SPECIFICATION

CUTTER, ORTHOPEDIC CAST, STRYKER, 110 Volt, AC-DC

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers an electric motor driven saw, with high speed oscillating drive, Stryker design, suitable for cutting surgical plaster casts.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-P-116	Preservation, Methods of.

FSC 6515

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

24160	Cutter, Orthopedic Cast, Stryker, 110 Volt, A.C.-D.C.

(Copies of Military specifications, standards and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, INC. (UL)

Underwriters' Laboratories Standards

Underwriters' Laboratories Standards for Safety

UL 45	Electric Tools

(Application for copies should be addressed to the Underwriters' Laboratories Inc., Walt Whitman Road, Melville, L.I., N.Y., 11746; 207 East Ohio Street, Chicago, Illinois, 60611; and 1655 Scott Boulevard, Santa Clara, California, 95050.)

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AMERICAN NATIONAL STANDARDS INSTITUTE (A. N. S. I.)

A. N. S. I. C73.11-1963 Outlet Receptacles, Attachment Plug Caps and Appliance Plugs.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, N. Y., N. Y.)

3. REQUIREMENTS

3.1 Fire and Casualty hazards.

3.1.1 The contractor shall submit to the contracting agency proof, prior to delivery, that the equipment supplied under this specification conforms to the applicable requirements of Underwriters' Laboratories Inc. Standard UL 45. Such proof shall consist of a laboratory report or listing by the Underwriters' Laboratories, Inc. The label or listing of the Underwriters' Laboratories shall be acceptable as evidence that the equipment conforms to the above. A copy of each laboratory report shall be furnished to the contracting officer.

3.1.2 In lieu of the label or listing by the Underwriters' Laboratories Inc., the contractor may submit a laboratory report by a nationally recognized testing agency adequately equipped and competent to perform such services, as shall be evidenced by the attachment of its seal, stamp or label to the article, or the issuance of a certificate. The contractor shall submit the laboratory report which shall include test results and methods used to determine compliance, to the contracting officer for approval. The Government reserves the right to require a facilities report showing the qualifications of the testing laboratory and to conduct a facilities survey to determine its acceptability.

3.1.3 Compliance with the above requirements regarding fire and casualty hazards, does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

3.2 Material. Material shall be of the highest quality throughout and suitable for the purpose intended.

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3.3 Style, design and dimensions. Shall be an electric motor driven cast cutter of the Stryker design, suitable for cutting casts on the human body without damage or injury to patient. Cast cutter shall be a nonrotating, high frequency, oscillating circular saw, complete with plug adapter, wrench, two 2 inch and two 2-1/2 inch diameter circular blades. Style, design and dimensions shall be in accordance with DPSC Drawing 24160, revised 19 December 1969.

3.3.1 Motor and switch. Motor shall operate on either 110 volt direct current or 110 volt, single phase, 60 cycles per second, alternating current. Motor shall be fan cooled, permanently lubricated, universal type, rated at 0.1 horsepower at 10,000 revolutions per minute continuous. Brush holders shall be protected by machined metal plugs threaded into and flush with cast cutter housing. Shall have "off" and "on" switch mounted on motor. Motor and switch shall comply with the requirements of Underwriters' Laboratories, Inc. Standard UL 45 for continuous rated universal motors for electric tools.

3.3.2 Power cable and plug. Cast cutter shall be furnished with a 3-wire, electrical power cord, 15 feet $\pm$ 1/2 foot long, and 3-prong plug in accordance with A. N. S. I. Standard C73.11. The power cable shall be provided with a spring type component to assure strain relief of power cable at the housing end.

3.3.3 Blade arbor. The centerline of the blade arbor shall be offset 1 $\pm$ 1/8 inch from the centerline of the motor shaft, as shown on drawing, to provide blade in a safe and convenient position for cast cutter operation. The arbor shall be supported in bearing sleeves on each side of mechanism (open link) that transmits rotating motion into oscillatory motion (see 3.3.5). Blade arbor shall be designed to provide a proper and secure fit for the blades specified in 3.3.7.

3.3.4 Center of balance. With supply cable supported or detached, the center of balance of the cast cutter shall be 6-5/16 $\pm$ 1/2 inches from the front side of the mounted 2-1/2 inch diameter blade (measured parallel to the longitudinal axis of the housing).

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3.3.5 Drive mechanism. The rotary power of the motor shall be transmitted into an oscillatory action, at the blade end, by means of a cam and open link mechanism. The cam shall be equipped with a roller bearing. The open link mechanism shall be mechanically fastened to the offset blade arbor (see 3.3.3) to suitably transmit required oscillation rate. Gear transmission design shall not be acceptable.

3.3.6 Housing. The working mechanism shall be enclosed by a compact, rigid, and thoroughly protected corrosion-resistant housing. The housing, except for the ventilating openings, shall totally enclose all stationary and rotating electrical parts. Six inlet air ports and four internal air circulation ports shall be provided in the housing for adequate ventilation. Four inlet air ports shall be provided at the brush end of the motor housing. Air shall be exhausted through eight air ports suitably sized at the front end of the motor housing. All external fastening devices shall be recessed into the housing. Contours of mating housing parts shall match, with no sharp corners or edges. Housing components shall be accurately fitted with adequate oil seals and gaskets to prevent oil leakage.

3.3.7 Blades. Each cast cutter shall be furnished with two, 2 inch and two, 2-1/2 inch diameter circular blades. The 2 inch and 2-1/2 inch diameter blades shall be in accordance with Purchase Descriptions for FSN 6515-323-4515 Blade, Orthopedic Cast Cutter, 2 inch Diameter and FSN 6515-323-4520 Blade, Orthopedic Cast Cutter, 2-1/2 inch Diameter, in effect on the date of invitations for bid.

3.3.8 Cast cutter weight. The cast cutter, including power cable, 3-prong plug and 2-1/2 inch diameter blade, shall not exceed 4-1/2 pounds (less carrying case, wrench and remaining blades).

3.3.9 Fastening devices. All screws, pins, bolts and similar parts shall be installed with means of preventing loss of tightness.

3.3.10 Wrench. Each cast cutter shall be furnished with a wrench, as shown on drawing, for attaching blades to arbor. Wrench shall be fabricated from corrosion-resistant metal.

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3.3.11 Cast cutter performance. With input voltage set at 115 volts, the unit shall be capable of meeting the following performance characteristics:

Characteristic	Requirement
No load oscillations (strokes) per minute (Free running)	17,000 minimum
Full load oscillations per minute under full load current of 1.2 amperes	13,000 minimum
No load current (Free running)	0.7 ampere maximum
Temperature rise of motor housing during 5 minute operating period, under full load current of 1.2 amperes (when operated from a cold start)	40 Degrees Fahrenheit maximum over ambient temperature
Continuous operation without breakdown or failure, under full load current of 1.2 amperes	100 hours minimum

3.3.12 Carrying case. The entire unit (including all accessories) shall be supplied in a rigid, carrying case. The carrying case shall be fitted with contoured cushioning material which shall secure all components during transit.

3.3.13 Service data. The contractor shall enclose, with each unit, two copies each of one or more booklets which contain complete instructions for installation, operation, maintenance, and a list of component parts. Wiring diagrams shall be included for electrical items or components. Each part of the unit other than common fasteners, etc., shall be identified in the service data and parts lists by the manufacturer's part number. As soon as practical after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by the Defense Personnel Support Center, Directorate of Medical Materiel. The cover of each service manual should indicate the following:

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FSN 6515-323-4510
CUTTER, ORTHOPEDIC CAST, STRYKER
Contractor's Trademark, Model Number, Name and Address
110 Volts, AC - DC

The preliminary service data shall be submitted sufficiently early to permit adequate review by the Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract, and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies of the approved service data may be waived upon request.

3.3.14 Identification marking. Each cast cutter shall be identified with a nameplate located on the motor housing, bearing the name or registered trademark of manufacturer, manufacturer's address, type or model number, serial number, voltage, full load current and cycle to be used for its operation. The top surface of the case shall be marked in black or gold with the letters "U. S." and the Federal stock number of the item. A metal plate, permanently die-stamped or etched with the manufacturer's name or registered trademark, shall be secured to the inside surface of the case cover.

3.3.15 Workmanship. Workmanship shall be first class throughout. Cast cutters shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the

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specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliance or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. Certificates of quality supplied by the manufacturer of the cast cutter may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the component, the lot number, the lot size, the sample size, the date of testing, the test method, the individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in Table I.

Table I. Sampling Criteria for Examination

Type Defect	Inspection Level *	AQL (percent defective)
Major	S-4	1.0
Minor	S-4	6.5

* If acceptance number is not exceeded, lot shall be acceptable. If rejection number is equalled or exceeded, then the inspection at level II may be performed to determine the final rejection or acceptance.

4.2.2 For tests.

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4.2.2.1 For performance. Sample size for performance of the cast cutters shall be as indicated for inspection level S-3 in MIL-STD-105, except that the acceptance number shall be zero for all sample sizes, and a minimum of two samples shall be taken.

4.2.2.2 For other tests. Sample size for each test (except for blades) shall be as indicated for inspection level S-1 in MIL-STD-105, except that the acceptance number shall be zero for all sample sizes, and a minimum of two samples shall be taken regardless of lot size. Sample size for tests pertaining to cast cutter blades shall be in accordance with those tests specified in applicable purchase description (see 3.3.7).

4.2.3 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The orthopedic cast cutter shall be examined to determine compliance with specification requirements. Examination shall be conducted as specified in Table II. Examination, however, shall not be restricted to defects listed in Tables I and II. Any cast cutter including cast cutter blades and other specified components containing one or more defects shall be rejected, and, if the number of defective items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table II. Classification of Defects

Categories	Defects
Critical	None defined.
Major:	
101	Evidence of Underwriters' Laboratories approval not obtained as indicated in 3.1.
102	Blades not furnished as specified in 3.3.7.

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Table II. Classification of Defects (Cont'd.)

Categories	Defects
Major:	
103	Switch not as specified.
104	Power cable and plug not as specified.
105	Power cable not reinforced at junction with tool or not type specified.
106	Wrench not as specified.
107	Case not as specified.
108	Evidence of lubricant leakage.
109	Housing ventilation not as specified.
110	Identification plates not as specified.
Minor:	
201	Instruction books and service data missing or not as specified.

4.4 Tests. Tests shall be conducted as necessary, to determine compliance with specification requirements.

4.4.1 Motor test. Each sample cast cutter motor shall be tested to determine compliance with Underwriters' Laboratories, Inc. Standard UL 45 requirements for continuous rated universal motors for electric tools for:

a. Continuity of grounding connections.
b. Dielectric strength.
c. Insulation resistance.
d. Current input.
e. Temperature rise.
f. Current leakage.

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4.4.2 Performance test. Sample cast cutters shall be tested for the requirements specified herein and as listed on identification plate:

a. Amperes at full load and no load on 115 volts, 60 cycles, alternating current.
b. Spindle speed, oscillations per minute at no load and full load.
c. Horsepower output.
d. Breakdown time (in hours) when under full load current.

4.4.2.1 General performance tests. After cast cutters have been tested as specified in 4.4.2, the cast cutter motors shall be checked for switch operation, excessive noise and vibration.

5. PREPARATION FOR DELIVERY.

5.1 Unit of issue. One cutter, one wrench, two blades and one carrying case, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table III shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table III. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	4 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Unit package. Each cutter along with component parts shall be positioned in their respective compartments within the carrying case. The complete unit shall be further packaged in accordance with MIL-P-116, method IA-15.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking. In addition to any special marking required by the contract or order, unit packages and exterior containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. This cast cutter is to be used for cutting casts on the human body without injury or damage to the patient.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Federal stock number.

6.3 This specification covers the following item appearing the Federal Supply Catalog:

FEDERAL STOCK NO.	ITEM IDENTIFICATION
6515-323-4510	Cutter, Orthopedic Cast, Stryker, 110 Volt, AC-DC

It does not include all types as indicated by the title nor all types commercially available.

6.4 Supercession data. This specification includes the requirements of Purchase Description No. 10, dated 6 January 1969.

Custodians:
Army - MD
Navy - MS
Air Force. - 03

Preparing activity:
Defense Supply Agency. - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document draft circulation should be based on the on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document-(s) or serve to amend contractual requirements .

SPECIFICATION

MIL-C-36625 FSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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